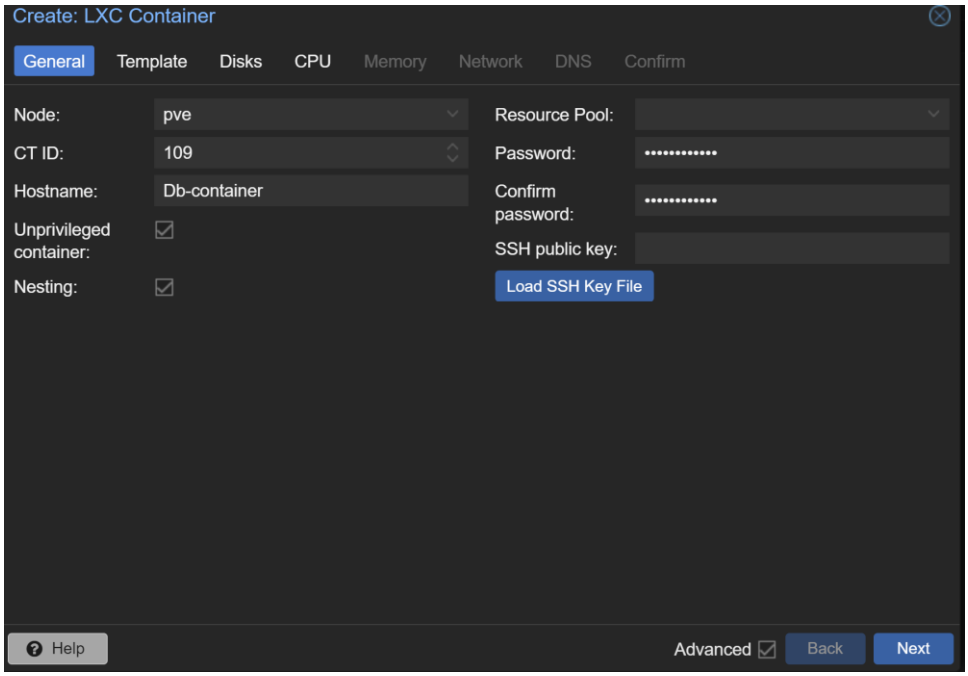
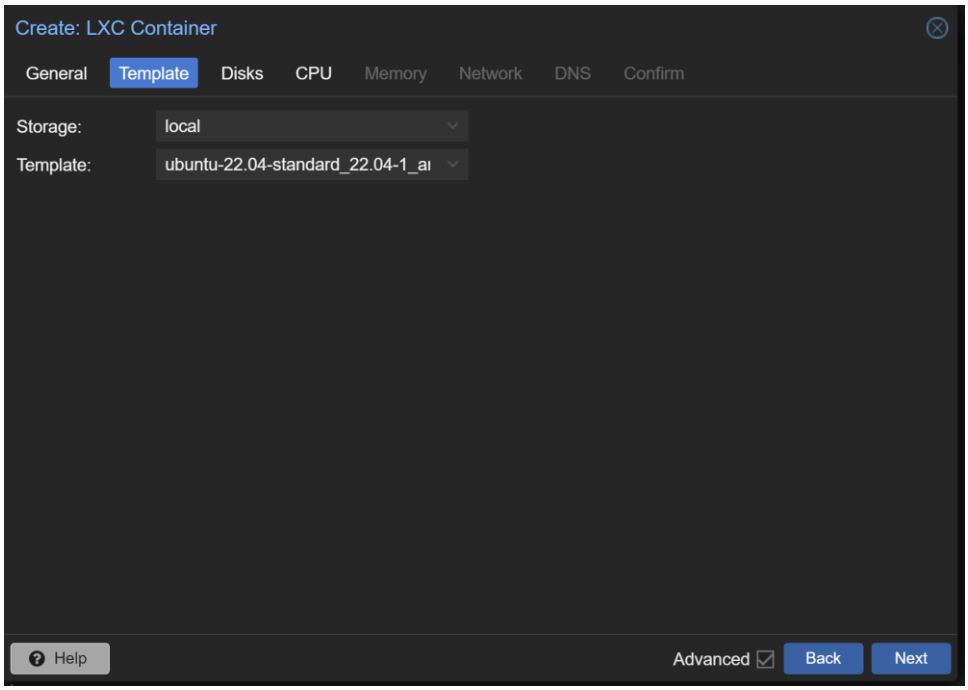
Création d’un conteneur pour centraliser la gestion des données de notre infrastructure :

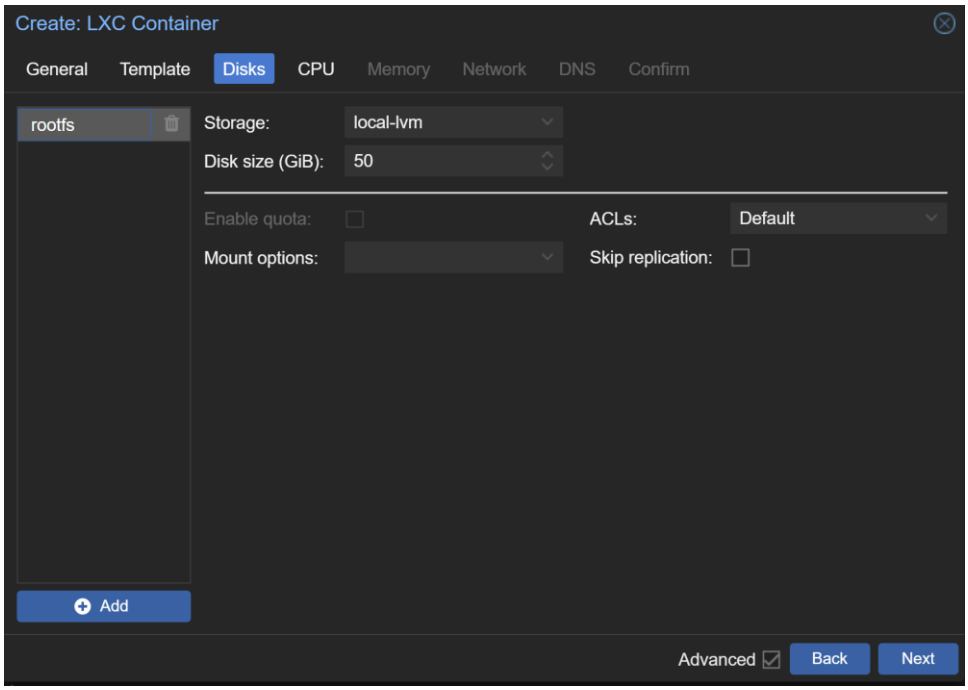
Création du conteneur qui va servir de stockage et de centralisation de nos données :



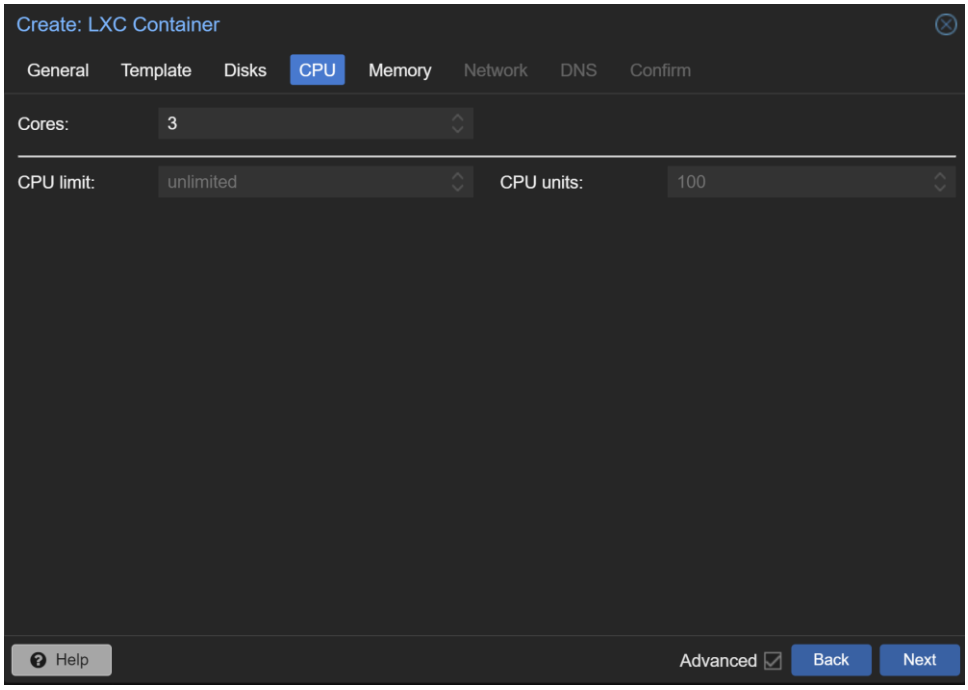
Choix de l’os du conteneur, dans notre cas “Ubuntu” sans interfaces graphique :



Attribution de l’espace disque, nous allons mettre 50 GO pour commencer, et si besoin est nous rajouterons un disque de stockage plus tard :



Attribution du nombre de cœur pour la base de données, 3 sont largement suffisants :



Une fois l’installation effectuée, on modifie le fichie de résolution DNS sur notre machine :

```
#  any later version.
#
# Entries in this file show the compile time defaults. Local configuration
# should be created by either modifying this file, or by creating "drop-ins" in
# the resolved.conf.d/ subdirectory. The latter is generally recommended.
# Defaults can be restored by simply deleting this file and all drop-ins.
#
# Use 'systemd-analyze cat-config systemd/resolved.conf' to display the full config.
#
# See resolved.conf(5) for details.

[Resolve]
# Some examples of DNS servers which may be used for DNS= and FallbackDNS=:
# Cloudflare: 1.1.1.1#cloudflare-dns.com 1.0.0.1#cloudflare-dns.com 2606:4700:4700::1111#cloudflare-dns.com 2606:
# Google:     8.8.8.8#dns.google 8.8.4.4#dns.google 2001:4860:4860::8888#dns.google 2001:4860:4860::8844#dns.goog
# Quad9:      9.9.9.9#dns.quad9.net 149.112.112.112#dns.quad9.net 2620:fe::fe#dns.quad9.net 2620:fe::9#dns.quad9.
DNS=8.8.8.8 8.8.4.4
FallbackDNS=8.8.8.8 8.8.4.4
#Domains=
#DNSSEC=no
#DNSOverTLS=no
#MulticastDNS=no
#LLMNR=no
#Cache=no-negative
#CacheFromLocalhost=no
#DNSStubListener=yes
#DNSStubListenerExtra=
#ReadEtcHosts=yes
#ResolveUnicastSingleLabel=no
```

Une fois que c’est fait et qu’on a configuré le DNS de google, on met à jour le systeme avec la commande “sudo apt update && sudo apt upgrade –y":

```
root@Db-container:~# sudo apt update && sudo apt upgrade -y
Get:1 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main Translation-en [510 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 c-n-f Metadata [30.3 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/restricted Translation-en [18.6 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 c-n-f Metadata [488 B]
Get:8 http://archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [1325 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [267 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [16.1 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [1342 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [220 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [520 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1042 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [235 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [22.1 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [42.1 kB]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [10.1 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [472 B]
Get:24 http://archive.ubuntu.com/ubuntu jammy-security/main amd64 Packages [1109 kB]
Get:25 http://archive.ubuntu.com/ubuntu jammy-security/main Translation-en [207 kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [11.4 kB]
Get:27 http://archive.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [1313 kB]
Get:28 http://archive.ubuntu.com/ubuntu jammy-security/restricted Translation-en [214 kB]
Get:29 http://archive.ubuntu.com/ubuntu jammy-security/restricted amd64 c-n-f Metadata [520 B]
Get:30 http://archive.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [837 kB]
Get:31 http://archive.ubuntu.com/ubuntu jammy-security/universe Translation-en [160 kB]
Get:32 http://archive.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [16.8 kB]
Get:33 http://archive.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [37.1 kB]
```

Une fois que c’est fais, on procède à l’installation de MariaDB à l’aide de la commande : sudo apt install mariadb-server -y

```
root@Db-container:~# sudo apt install mariadb-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  galera-4 gawk libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl libdaxctl1
  libdbd-mysql-perl libdbi-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0ldbl libgdbm-compat4
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmariadb3 libmpfr6 libmysqlclient21 libndctl6 libperl5.34 libpmem1
  libsigsegv2 libsnappylv5 libterm-readkey-perl libtimedate-perl liburi-perl liburing2 mariadb-client-10.6
  mariadb-client-core-10.6 mariadb-common mariadb-server-10.6 mariadb-server-core-10.6 mysql-common perl
  perl-modules-5.34 socat
Suggested packages:
  gawk-doc libmldbm-perl libnet-daemon-perl libsql-statement-perl libdata-dump-perl libipc-sharedcache-perl
  libbusiness-isbn-perl libwww-perl mailx mariadb-test perl-doc libterm-readline-gnu-perl
  | libterm-readline-perl-perl make libtap-harness-archive-perl
The following NEW packages will be installed:
  galera-4 gawk libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl libdaxctl1
  libdbd-mysql-perl libdbi-perl libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0ldbl libgdbm-compat4
  libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmariadb3 libmpfr6 libmysqlclient21 libndctl6 libperl5.34 libpmem1
  libsigsegv2 libsnappylv5 libterm-readkey-perl libtimedate-perl liburi-perl liburing2 mariadb-client-10.6
  mariadb-client-core-10.6 mariadb-common mariadb-server mariadb-server-10.6 mariadb-server-core-10.6
  mysql-common perl perl-modules-5.34 socat
0 upgraded, 43 newly installed, 0 to remove and 0 not upgraded.
Need to get 28.6 MB of archives.
After this operation, 218 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmpfr6 amd64 4.1.0-3build3 [1425 kB]
0% [1 libmpfr6 14.2 kB/1425 kB 1%]
```

Une fois que c’est effectué, on va lancer le script de sécurisation de maria db à l’aide de la commande “sudo mysql_secure_installation” :

```
root@Db-container:~# sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
 ... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] n
 ... skipping.

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] Y
 ... Success!
```

Sécurisez la base de données comme vous le souhaitez, vous devez avoir le message suivant s’afficher :

```
All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
```